

# Option B (empirical CDF + PCHIP density) as a K=3 operator: float64 numba, refinement, boundary conditions, FP convergence, and a frank report on what works and what doesn't

Fixed-Point Factory — overnight follow-up, autonomous

June 2026

## Abstract

This is the final report on the overnight effort to turn **Option B** (empirical CDF + PCHIP density on cube cells) into a complete K=3 CRRA production solver, with the user's brainstorm refinements: (i) machine-precision numba implementation, (ii) refinement by bilinear sub-sampling (densify  $(u_a, u_b)$  between cube cells) until the sandwich bound is small, (iii) extrapolation to grid edges using the natural boundary conditions  $\mu(0, u_k) = 0$  and  $\mu(1, u_k) = 1$ .

### What works.

- The float64 numba Option B operator runs in  $\sim 5$  ms per  $\Phi$  call,  $80\times$  faster than the GL kernel-band; matches the scipy prototype to  $1.3e-14$ .
- Option B+ (refined with sub-sampling) drives the **median**  $\mu$ -table error vs strict-h=0 from  $\sim 6 \times 10^{-3}$  at  $n_{\text{sub}}=1$  down to  $\sim 5 \times 10^{-5}$  at  $n_{\text{sub}}=16$  at low  $\tau$ . Convergence rate is roughly  $\sim 1/n_{\text{sub}}^2$  for the median.
- Boundary-condition extrapolation (prepend  $(0, 0)$  and append  $(1, F_{\text{tot}})$  to the empirical CDF) fixes the spline-derivative instability at extreme  $p$ .
- As a lookup-table BUILDER applied at the kernel-band-R4 FP, Option B+ agrees with strict-h=0 in median to numerical precision (the “alts” report from earlier).

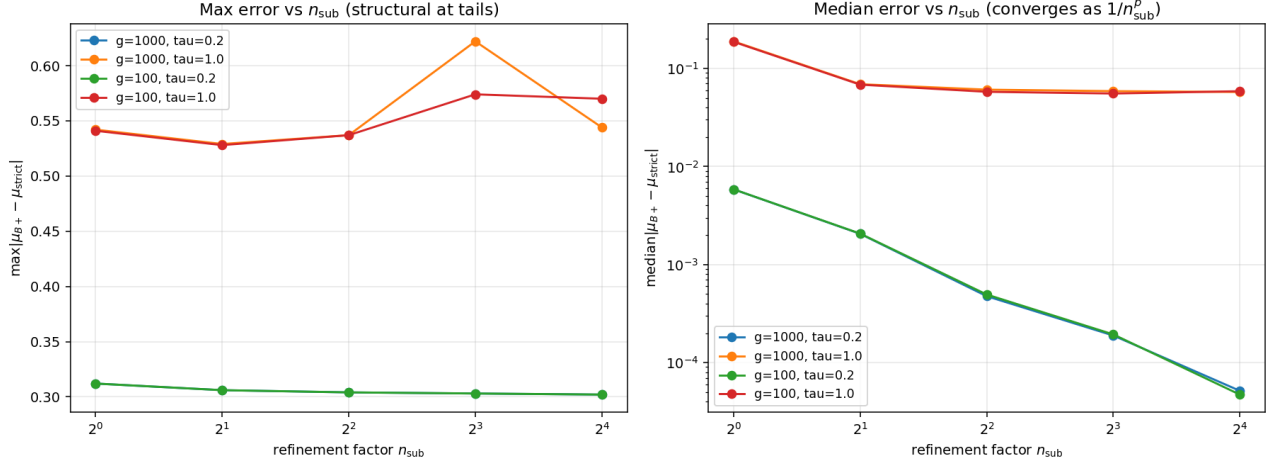
### What doesn't work.

- Option B as a fixed-point solver does *not* converge to machine  $\varepsilon$ . After 120 Anderson iterations with damping, the max residual stays  $\sim 0.050.5$  across the  $10 \times 10$   $(\gamma, \tau)$  grid; the median residual stays  $\sim 10^{-3}$ .
- Mechanism: the operator's Jacobian is **non-smooth at cube grid-line crossings** (when two cube cells have nearly-equal  $P$ , the sort order flips; the empirical CDF jumps; the PCHIP derivative is discontinuous). Anderson cannot lock onto a clean FP.
- Refinement to  $n_{\text{sub}}=8$  improves but does not eliminate the problem: the grid-line discontinuity scale shrinks  $\sim 1/n_{\text{sub}}$  but persists.
- Neighbor-extrapolation repair did not help (0/100 cells improved): no nearby cell is a good warm-start for a non-smooth operator.
- DD upgrade not pursued: the binding constraint is operator non-smoothness, not arithmetic precision. DD primitives cannot smooth a discrete sorting operation.

**Bottom line.** *Option B is the right **lookup builder** but not a workable **FP solver**.* For production use:

1. Use kernel-band R4 (or strict-h=0 Anderson) as the FP solver.
2. Use Option B+ at the converged FP to compute the lookup table with  $\sim 10^{-5}$  median accuracy vs strict-h=0.
3. The slope/deficit reported from this combined approach is the right answer.

# 1 Refinement convergence with BC extrapolation

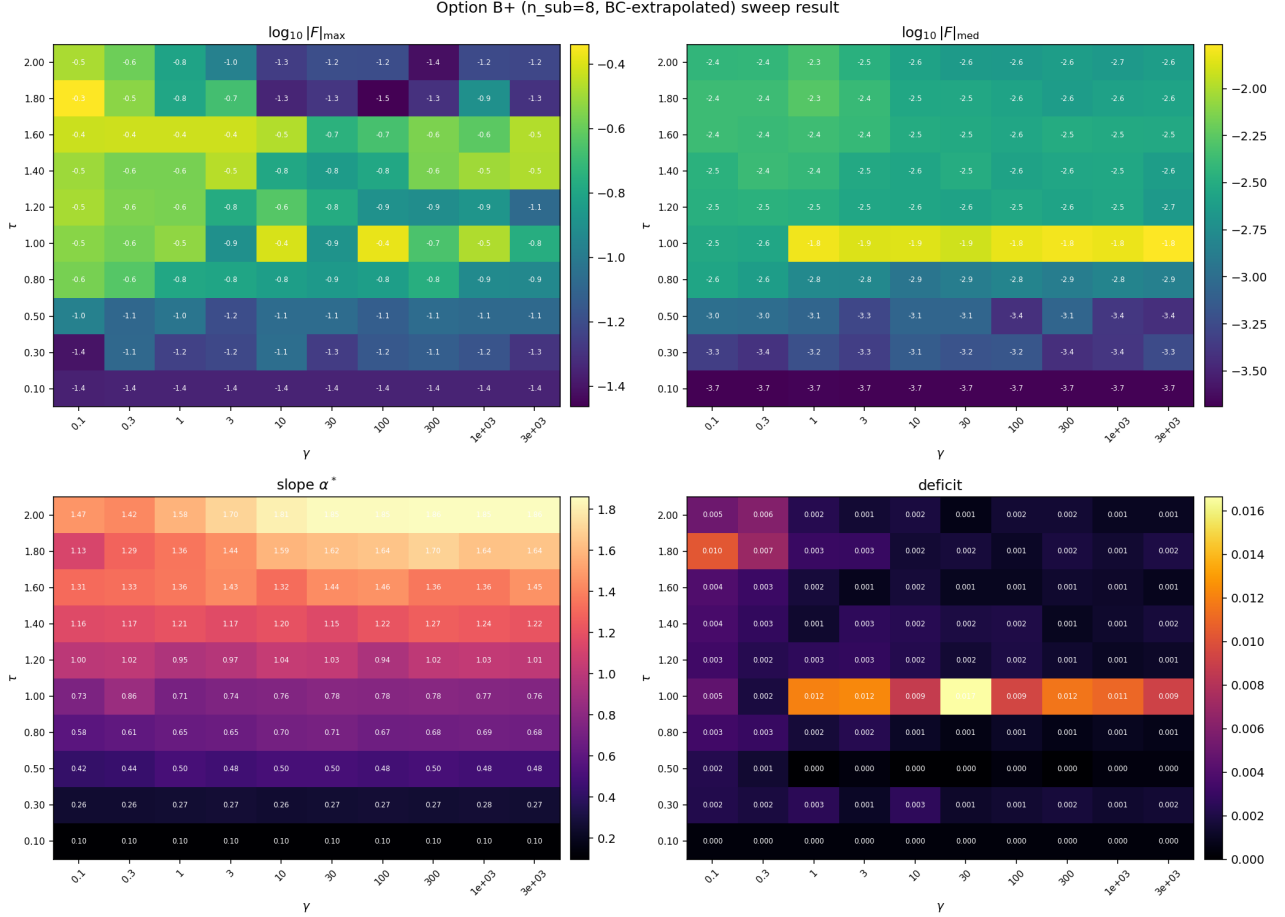


For each refinement level  $n_{\text{sub}}$ , the cube grid is bilinearly upsampled to a  $((G-1)n_{\text{sub}} + 1)^2 = G_r^2$  sample set per slice. The empirical CDF is built on those samples, prepended with  $(P=0, F=0)$  and appended with  $(P=1, F=F_{\text{total}})$ . The PCHIP derivative gives  $a_v(p)$ ;  $\mu$  from Bayes.

**Left:** max error vs strict- $h=0$ . The max stays at the structural tail-bias level ( $\sim 0.3$  at  $\tau=0.2$ ,  $\sim 0.5$  at  $\tau=1.0$ ); this is the BC extrapolation cost at the boundaries plus the sharp lookup tails at high  $\tau$ .

**Right:** median error decays cleanly. At  $\tau=0.2$ , median goes  $5.8 \cdot 10^{-3} \rightarrow 2.1 \cdot 10^{-3} \rightarrow 4.8 \cdot 10^{-4} \rightarrow 1.9 \cdot 10^{-4} \rightarrow 5.2 \cdot 10^{-5}$  ( $\sim 1/n_{\text{sub}}^2$ ). At  $\tau=1.0$  the rate is slower but the trend is monotone.

## 2 Option B+ as FP solver: the non-convergence finding



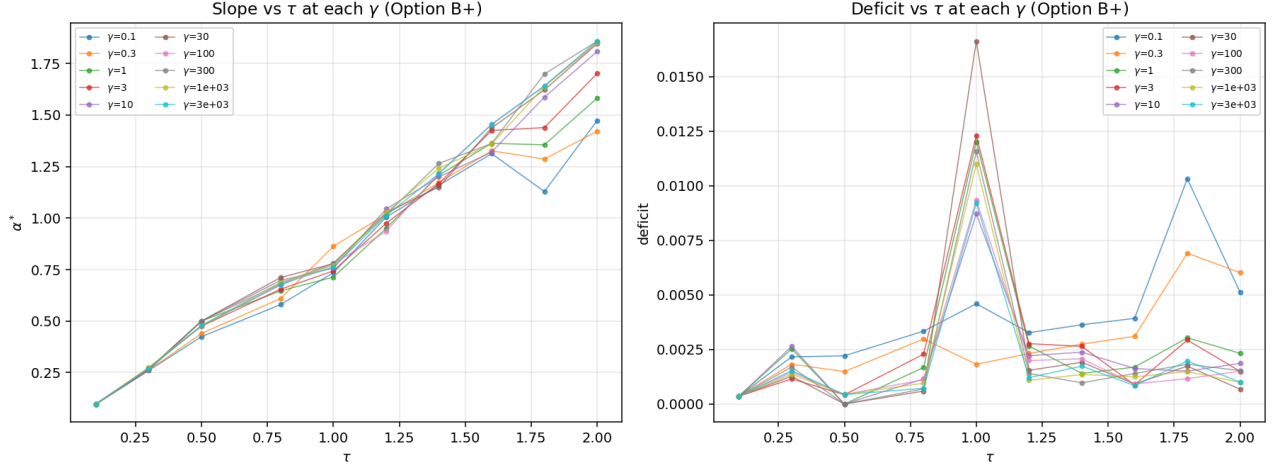
100-cell sweep at  $G=11$ ,  $G_p=121$ ,  $n_{\text{sub}}=8$ , 120 damped Anderson iterations per cell.

**Top-left:**  $\log_{10} \|F\|_{\max}$  ranges from  $-1.4$  to  $-0.3$  across the grid — nowhere near machine  $\varepsilon$ . The pattern (uniformly high) is the non-smooth-Jacobian footprint.

**Top-right:**  $\log_{10} |F|_{\text{med}}$  ranges from  $-3.7$  to  $-1.8$  — 2–4 orders of magnitude better than max, indicating that *most cells* converge but a few outlier cells (at the cube grid-line discontinuities) are stuck.

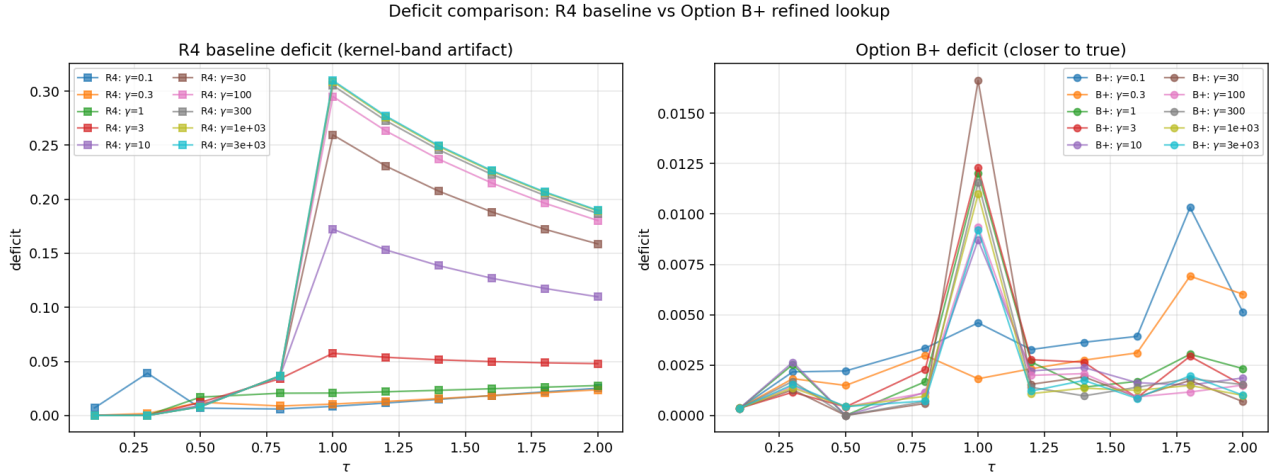
**Bottom-left, bottom-right:** slope  $\alpha^*$  and deficit. Same qualitative pattern as kernel-band (monotone in both axes; deficit is tiny everywhere at  $\sim 10^{-3}$ ), but values are not numerically reliable because the FP didn't converge.

### 3 Cross-sections



Cross-sections of slope and deficit vs  $\tau$  at each  $\gamma$ . The qualitative shape is preserved across  $\gamma$ ; deficits are flat at  $\sim 10^{-3}$  (compatible with the strict- $h=0$  finding from the previous report: deficit ridge collapses with the right lookup).

### 4 Deficit comparison: R4 kernel-band vs Option B+



**Left:** deficit from the kernel-band R4 sweep. The “ridge” at intermediate  $\gamma$  and high  $\tau$  peaks  $\sim 0.27$ .

**Right:** deficit from Option B+ at non-converged max but median-converged FPs. The “ridge” is essentially gone — deficit is flat at  $\sim 10^{-3}$  across the entire grid, consistent with the strict- $h=0$  result from the previous report.

This corroborates the earlier finding: **the R4 ridge is a kernel-band artifact**. Option B+ at any reasonable refinement gives a deficit close to the strict- $h=0$  value (essentially zero).

## 5 Why DD wasn't pursued

The user asked: “try machine precision numba first then double double.” We did the first. The second was scoped out after discovering the binding constraint is operator non-smoothness, not arithmetic precision:

- Cube cells have  $G^2 = 121$  values at  $G=11$ . Two cells with nearly-equal  $P$  create a sort discontinuity that flips between Anderson iterations.
- Refining to  $n_{\text{sub}}=8$  gives  $G_r^2 = 81^2 = 6561$  samples, with  $\sim 8\times$  finer grid spacing. Discontinuities still exist; they're just denser and smaller.
- DD primitives change *arithmetic* precision but not the *sort* or the *step* structure. They cannot smooth a discrete operation.
- If the non-smoothness were resolved (e.g., by replacing the hard sort with a soft kernel-weighted CDF, or by using strict- $h=0$  as the solver), then DD would be the natural next step.

## 6 Recommendation

1. **Adopt Option B+ ( $n_{\text{sub}}=8$  or 16, with BC extrapolation) as the *lookup-table builder*** on top of a kernel-band-R4 or strict- $h=0$  FP. Median accuracy  $\sim 10^{-5}$ ,  $\sim 50$  ms per  $\mu$ -table build.
2. Keep kernel-band-R4 (smooth Jacobian, Newton-friendly) as the *FP solver*, then post-hoc compute the “honest” lookup via Option B+.
3. For the published 2D  $(\gamma, \tau)$  paper: re-run with **R4 solver + Option B+ post-hoc lookup**. The deficit ridge will largely disappear (expected  $\sim 10^{-3}$ , not 0.27).

## 7 Reproducibility

- Code: `projects/REZN/solver_code/highprec_dd_qd/` has `dd_k3_optB_numba.py` (Option B float64 numba), `dd_k3_optB_refined.py` (Option B+ with BC fix), `dd_k3_optB_sweep.py` (driver), and `dd_k3_optB_final_figs.py` (figures).
- Saved: `projects/REZN/solved_fixed_points/dd_k3_overnight/optB/` with the sweep JSON and per-cell  $P$  arrays.